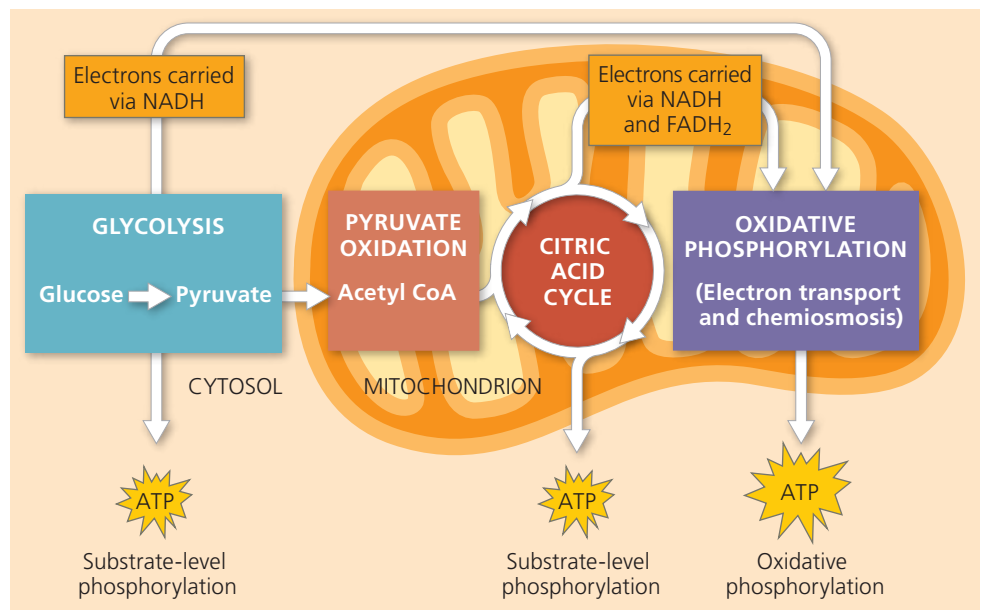


► **Figure 9.5 An overview of cellular respiration.** During glycolysis, each glucose molecule is broken down into two molecules of pyruvate. In eukaryotic cells, as shown here, the pyruvate enters the mitochondrion. There it is oxidized to acetyl CoA, which will be further oxidized to CO₂ in the citric acid cycle. The electron carriers NADH and FADH₂ transfer electrons derived from glucose to electron transport chains. During oxidative phosphorylation, electron transport chains convert the chemical energy to a form used for ATP synthesis in the process called chemiosmosis. (During earlier steps of cellular respiration, a few molecules of ATP are synthesized in a process called substrate-level phosphorylation.) To visualize these processes in their cellular context, see Figure 6.32b.

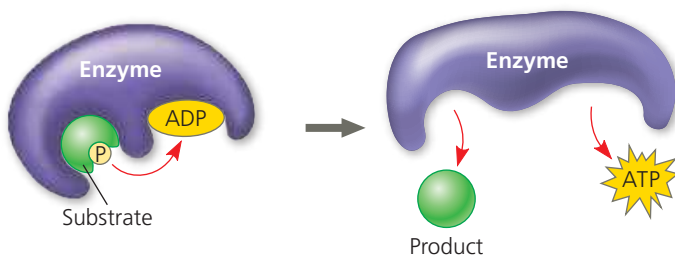
➔ **Mastering Biology Animation: Overview of Cellular Respiration**



electrons from NADH or FADH₂ generated during the first two stages and passes these electrons down the chain. At the end of the chain, the electrons are combined with molecular oxygen (O₂) and hydrogen ions (H⁺), forming water (see Figure 9.4b). The energy released at each step of the chain is stored in a form the mitochondrion (or prokaryotic cell) can use to make ATP from ADP. This mode of ATP synthesis is called **oxidative phosphorylation** because it is powered by the redox reactions of the electron transport chain.

In eukaryotic cells, the inner membrane of the mitochondrion is the site of electron transport and another process called *chemiosmosis*, together making up oxidative phosphorylation. (In prokaryotes, these processes take place in the plasma membrane.) Oxidative phosphorylation accounts for almost 90% of the ATP generated by respiration. A smaller amount of ATP is formed directly in a few reactions of glycolysis and the citric acid cycle by a mechanism called **substrate-level phosphorylation** (Figure 9.6). This mode of ATP synthesis

▼ **Figure 9.6 Substrate-level phosphorylation.** Some ATP is made by direct transfer of a phosphate group from an organic substrate to ADP by an enzyme. (For examples in glycolysis, see Figure 9.8, steps 7 and 10.)



MAKE CONNECTIONS Review Figure 8.9. In the reaction shown above, is the potential energy higher for the reactants or for the products? Explain.

occurs when an enzyme transfers a phosphate group from a substrate molecule to ADP, rather than adding an inorganic phosphate to ADP as in oxidative phosphorylation. “Substrate molecule” here refers to an organic molecule generated as an intermediate during the catabolism of glucose. You’ll see examples of substrate-level phosphorylation later in the chapter, in both glycolysis and the citric acid cycle.

You can think of the whole process this way: When you withdraw a relatively large sum of money from an ATM, it is not delivered to you in a single bill of a large denomination. Instead, the machine dispenses a number of smaller-denomination bills that you can spend more easily. This is analogous to ATP production during cellular respiration. For each molecule of glucose degraded to CO₂ and H₂O by respiration, the cell makes up to about 32 molecules of ATP, each with 7.3 kcal/mol of free energy. Respiration cashes in the large denomination of energy banked in a single molecule of glucose (686 kcal/mol under standard conditions) for the small change of many molecules of ATP, which is more practical for the cell to spend on its work.

This preview has introduced you to how glycolysis, the citric acid cycle, and oxidative phosphorylation fit into the process of cellular respiration so you can keep the big picture in mind as you take a closer look at each of these three stages of respiration. As you read about the chemical reactions, remember that each reaction is catalyzed by a specific enzyme, some of which are shown in Figure 6.32b.

CONCEPT CHECK 9.1

1. Compare and contrast aerobic and anaerobic respiration, including the processes involved.
2. **WHAT IF?** If the following redox reaction occurred, which compounds would be oxidized? Reduced?



For suggested answers, see Appendix A.